Manoj Kumar
Data Scientist & GenAI Engineer

## Experience Summary

- 8 years of experience in Data Science, Machine Learning, MLOPs and Gen AI.
- Conducted comprehensive data preparation and feature engineering, ensuring high data quality and completeness for robust model performance.
- Development of AI and machine learning models to solve complex business problems and drive actionable insights and Hands on experience in developing Test Scripts.
- Applied knowledge of Transformer-based encode-decoder, decoder only architectures (BERT, RoBERTa, T5, GPT, Llama2) for innovative NLP solutions. Proficient in LLM model fine-tuning using LoRA and QLoRA enhancing its capabilities and RAG applications.
- Implemented Responsible and Ethical AI use cases like hallucination detection, semantic de-duplication, bias, and toxicity detection, Utilized Evidently AI for continuous model monitoring ( model drift, data drift, column drift, embedding drift, OOV) and performance assessment.
- Proficient in utilizing Azure cloud infra for development of Pipelines for creating and deploying models for business use cases through continuous integration and deployment (CI/CD) pipelines and MLOPs tools like MLFlow and MLFlow Recipe.

## Skills Summary

| | |
|---|---|
| Domain | Artificial Intelligence for IT Service Management (AI for ITSM), E-Learning, Energy Trading / Commodity Market Analytics, |
| Programming Languages | Python |
| Tools / DB / Packages / Framework | Machine Learning, Deep Learning: (ANN \| Pytorch), NLP: (RNN \| LSTM \| Encoder Decoder \| Attention \|Transformer \| BERT \| GPT), Generative AI: (Open AI, Llama, Gemini, Embeddings, HuggingFace, RAG, Finetuning, AI Agents), LLM Framework: (Langchain \| LangGraph \| LCEL \| LangSmith), Vector Databases: (FAISS \| ChromaDB \| Pinecone), RestAPIs: (FastAPI \| Flask), Databases: (MySQL \| MongoDB), MLOps: (GIT \| Docker \| Kubernetes \| DVC \| MLflow \| Kubeflow \| GitHub Actions) |
| Cloud Infra | AWS, Azure |

# Work Experience

## Project 1

| | | | |
|---|---|---|---|
| **Organization Name** | Coforge | Team Size | 4 |
| **Start Date** | Jan 2026 | End Date | Aug 2026 |
| **Project Name** | Froud Detection System | | |
| **Project Description** | Developed and optimized machine learning and Graph Neural Network (GNN)-based fraud detection solutions for the Value-Added Services (VAS) layer of the Fiserv payment gateway, enabling real-time transaction risk assessment and improving fraud detection performance across multiple enterprise clients.<br><br>Key Responsibilities:<br><br>• Developed and optimized fraud detection models using CatBoost and Graph Neural Networks (GNN) to identify suspicious transaction patterns and fraudulent activities.<br><br>• Implemented combined-model inference strategies integrating CatBoost and GNN architectures, leveraging graph-based relationships through Amazon Neptune.<br><br>• Performed feature engineering and dimensionality reduction using PCA, correlation analysis, and feature selection to improve model efficiency and prediction performance.<br><br>• Applied K-Modes clustering and undersampling techniques to handle data imbalance and improve training efficiency.<br><br>• Optimized fraud detection performance, achieving a 200% increase in True Positives, 66% reduction in False Positives, and 40% improvement in inference latency.<br><br>• Developed Docker-based model deployment pipelines and conducted scalability and stress testing with up to 1,000 concurrent users.<br><br>• Designed Multi-Model Endpoint (MME) and Multi-Variant Experimentation (MVE) strategies to optimize customer onboarding, model initialization, deployment scalability, and infrastructure costs.<br><br>• Delivered fraud detection solutions for enterprise clients, including Clover, The Home Depot, Disney, Wawa, Snappy, CardConnect, and CardPointe. | | |

| | |
|---|---|
| **Role & Contribution** | 1. Worked as a Developer for application development and<br>2. Implemented data quality systems to ensure high data integrity |
| **Technology &Tools** | Python, CatBoost, Graph Neural Networks (GNN),Amazon Neptune, PCA,K-Modes Clustering, Undersampling, Feature Engineering, Docker, Model Inference Optimization, MLOps, Stress Testing, AWS SageMaker |

## Project 2

| | | | |
|---|---|---|---|
| **Organization Name** | HCL | Team Size | 2 |
| **Start Date** | Jul 2025 | End Date | Oct 2025 |
| **Project Name** | Smart Service Desk | | |
| **Project Description** | Designed and developed an Agentic AI-powered Enterprise IT and Operations Service Desk application to automate employee support requests, streamline IT service management, and improve issue resolution through intelligent multi-agent orchestration and Retrieval-Augmented Generation (RAG).<br><br>Key Responsibilities:<br><br>• Designed and developed an end-to-end Agentic AI architecture using LangGraph, LangChain, Python, and LLMs to automate enterprise IT and Operations support workflows.<br><br>• Implemented intelligent intent classification and dynamic agent routing to identify employee requests, categorize issues, prioritize tasks, and execute specialized workflows.<br><br>• Developed RAG-based knowledge retrieval pipelines incorporating document ingestion, text chunking, embedding generation, vector databases, and semantic/hybrid search.<br><br>• Implemented LLM-powered contextual response generation to provide relevant troubleshooting solutions using enterprise FAQs, SOPs, and technical documentation.<br><br>• Built automated ticket management and escalation workflows to create support tickets, assign priorities, and route unresolved issues to appropriate L2/L3 support teams.<br><br>• Integrated enterprise systems and approval-based workflows to automate access requests, ticket updates, and email notifications. | | |

| | |
|---|---|
| **Role & Contribution** | 1. Worked as a Developer for application development and<br>2. Implemented data quality systems to ensure high data integrity |
| **Technology &Tools** | OpenAI, Pinecone, LangChain,LangGragh, Azure Document Intelligence, FastAPI, LangSmith, Docker, Kubernetes, Python |

## Project 3

| | | | |
|---|---|---|---|
| **Organization Name** | HCL | Team Size | 2 |
| **Start Date** | Jun 2025 | End Date | Jul 2025 |
| **Project Name** | ChatBot | | |
| **Project Description** | developed an intelligent chatbot powered by Retrieval-Augmented Generation (RAG) to assist users in querying detailed information about a specific enterprise application and its associated support workflows. The chatbot is capable of answering questions related to how support tickets are raised, categorized, escalated, and resolved across different operational teams, thereby improving user self-service and reducing dependency on manual support. | | |
| **Role & Contribution** | 3. Worked as a Developer for application development and<br>4. Implemented data quality systems to ensure high data integrity | | |
| **Technology &Tools** | OpenAI, Pinecone, LangChain, FastAPI, LangSmith, Docker, Kubernetes, Python | | |

## Project 4

| | | | |
|---|---|---|---|
| **Organization Name** | HCL | Team Size | 1 |
| **Start Date** | Apr 2025 | End Date | June 2025 |
| **Project Name** | Query Resolution Chatbot | | |
| **Project Description** | Developed an AI-driven chatbot leveraging Retrieval-Augmented Generation (RAG) to retrieve and deliver the most contextually relevant responses from a question–answer dataset. The system employs advanced natural language processing to understand user intent, match queries with precise answers, and generate accurate responses in real time. This solution enhances enterprise knowledge retrieval, automated customer support, and FAQ management, significantly improving query resolution efficiency and accuracy | | |
| **Role & Contribution** | 1. Worked as a Developer for development and<br>Implemented data quality systems to ensure high data integrity | | |
| **Technology &Tools** | OpenAI,Pinecone,LangChain,LangSmith,FastAPI ,Docker,Kubernetes | | |

## Project 5

| | | | |
|---|---|---|---|
| **Organization Name** | HCL | Team Size | 1 |
| **Start Date** | Jan 2025 | End Date | March 2025 |
| **Project Name** | Automated Ticket Categorization. | | |
| **Project Description** | Developed a deep learning–based multiclass text classification model to automatically categorize user-generated IT support messages (e.g., application access issues, attendance marking failures, app malfunctions) into predefined categories such as SSD, LMS, VSD, and HR. This solution assists users by recommending the appropriate category while raising tickets, improving routing accuracy and reducing manual effort. | | |
| **Role & Contribution** | Worked as a Developer, responsible for the end-to-end design and implementation. | | |
| **Technology &Tools** | BERT-base-uncased,LORA,PyTorch,Docker,Git,DVC,Kubernetes,Python | | |

## Project 6

| | | | |
|---|---|---|---|
| **Organization Name** | HCL | Team Size | 1 |
| **Start Date** | Oct 2024 | End Date | Dec 2024 |
| **Project Name** | Machine Translation | | |
| **Project Description** | Developed a bidirectional language translation model for English ↔ German using Hugging Face Transformers, MarianMT, TensorFlow, Python, and FastAPI. The model provides real-time, high-accuracy translations, making it ideal for applications in multilingual communication, content localization, and language learning. | | |
| **Role & Contribution** | Worked as a Developer, responsible for the end-to-end design, development, and deployment of the bidirectional English ↔ German translation model. | | |
| **Technology &Tools** | MarianMT (Helsinki-NLP),TensorFlow,PyTorch,FastAPI,Docker,Kubernetes,Python | | |

## Project 7

| | | | |
|---|---|---|---|
| **Organization Name** | NTPC | Team Size | 2 |
| **Start Date** | Jan 2021 | End Date | June 2021 |
| **Project Name** | Coal Auction Price Optimization | | |
| **Project Description** | Developed a predictive pricing model to align coal auction prices with real-time market demand, improving pricing accuracy and decision-making. Applied regression and time series techniques to forecast demand and recommend optimal starting bids. Enabled real-time insights for auction planning, reducing pricing inefficiencies and increasing buyer<br><br>participation. Designed the solution for scalability across mines and states with minimal customization. Delivered actionable intelligence dashboards for stakeholders to monitor demand trends and adjust strategy dynamically. | | |
| **Role & Contribution** | Worked as a Developer, responsible for dashboard design and model testing activities.<br><br>Designed interactive dashboards to visualize key performance indicators (KPIs) such as demand forecasts, auction prices, and participation trends for business stakeholders.<br><br>Tested and validated predictive models to ensure forecasting accuracy and model robustness under different market scenarios.<br><br>Ensured data consistency and performance optimization for real-time monitoring and reporting. | | |
| **Technology &Tools** | Python (FastAPI, XGBoost, SciPy), SQL, Docker/Kubernetes, AWS | | |

## Project 8

| | | | |
|---|---|---|---|
| **Organization Name** | NTPC | Team Size | 1 |
| **Start Date** | Jul 2021 | End Date | Sep 2021 |
| **Project Name** | Chatbot | | |
| **Project Description** | Designed and deployed a production-ready AI chatbot for NTPC to streamline customer inquiries related to its core business areas: Coal Mines, Coal Stations, Gas Stations, Renewable Energy, and Hydro Stations. The chatbot provides fast, accurate, and context-aware responses to user queries by integrating data directly from NTPC's public information sources. | | |

| | |
|---|---|
| **Role & Contribution** | Worked as a Developer, responsible for the end-to-end design, development, and deployment of a production-ready |
| **Technology &Tools** | GPT(OpenAI/LLM),Pinecone,LangChain,FastAPI,Docker,Kubernetes,Python |

## Project 9

| | | | |
|---|---|---|---|
| **Organization Name** | Wayinfotech Software Solutions Pvt. Ltd | Team Size | 2 |
| **Start Date** | Oct 2020 | End Date | Nov 2020 |
| **Project Name** | Sentiment analysis. | | |
| **Project Description** | Developed a Sentiment Analysis Model using TensorFlow, LSTM, and FastAPI to classify text as positive or negative. This model processes user-generated text and predicts the emotional tone using deep learning techniques. | | |
| **Role & Contribution** | Worked as a Developer, responsible for the end-to-end design, development, and deployment of the Sentiment Analysis Model. | | |
| **Technology &Tools** | Python, Azure Databricks. MLFlow Recipe | | |

## Organization: Woolworths

## Project 10

| | | | |
|---|---|---|---|
| **Organization Name** | Wayinfotech Software Solutions Pvt. Ltd. | Team Size | 2 |
| **Start Date** | Jan 2020 | End Date | Oct 2022 |
| **Project Name** | Multiple Choice Question Generator | | |
| **Project Description** | Developed an AI-powered multiple-choice question (MCQ) generation system that automatically creates objective questions — including multiple correct answers — from a given text or chapter. The solution enhances student engagement, promotes critical thinking, and assists educators in designing high-quality assessments efficiently with minimal manual effort | | |
| **Role & Contribution** | Worked as a Developer, responsible for the end-to-end design and development of the AI-powered MCQ generation system.<br><br>Collected, cleaned, and preprocessed textual data to create a structured dataset for training and evaluation. | | |

| | |
|---|---|
| | Developed REST APIs using FastAPI for real-time quiz generation and integration with e-learning platforms.<br><br>Containerized and deployed the solution using Docker/Kubernetes for scalability and efficient model serving.<br><br>Conducted model evaluation and optimization to ensure high-quality question generation and contextual accuracy. |
| **Technology &Tools** | Python,PyTorch,Sense2Vec,T5-Base Pretrained Summarizer,Pretrained T5 Question Generator,MSMARCO-DistilBERT-Base-V2,FastAPI |

### Project 11

| | | | |
|---|---|---|---|
| **Organization Name** | HCL | Team Size | 2 |
| **Start Date** | Jan 2022 | End Date | Feb 2022 |
| **Project Name** | Intelligent Customer Support Ticket Routing using NER, Intent Classification | | |
| **Project Description** | Developed an AI-driven customer support routing system that automatically classifies and assigns customer queries to the correct department (HR, Finance, IT, etc.) using a hybrid NLP pipeline. Integrated Named Entity Recognition (NER) for keyword extraction and a BERT-based intent classifier for contextual understanding. Implemented fusion logic to combine both signals. | | |
| **Role & Contribution** | End-to-End Solution Ownership: Designed, developed, and deployed a full-scale AI-driven customer support routing system capable of automatically classifying and assigning customer queries to the right department (HR, Finance, IT, etc.).<br><br>Hybrid NLP Pipeline: Engineered a hybrid pipeline combining Named Entity Recognition (NER) for entity extraction and a BERT-based intent classifier for contextual understanding.<br><br>Fusion Logic Implementation: Developed intelligent signal fusion logic to combine NER and intent classifier outputs, enhancing precision and recall for department classification. | | |
| **Technology &Tools** | Python, FastAPI, spaCy, BERT | | |

# Educational Qualification

| | |
|---|---|
| **Education & Credentials** | M.Sc. (Computer Applications) |